

Jenkins lab 1

Part 1 – Explore IDE and create a Jenkin job

Objectives

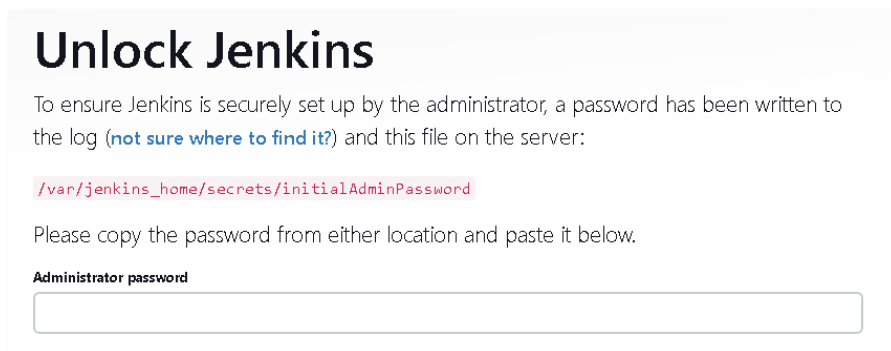
- In this part you will install Jenkins using Docker technology.
- Investigate the Jenkins IDE.
- Create a basic Job and explore its options

Install Jenkins using Docker

- For a simple method, type the following command

```
docker run -d -p 8080:8080 -v jenkins_home:/var/jenkins_home jenkins/jenkins:lts
```

- 1- Get the Jenkin's docker ID (**docker ps**).
In this example we assume the ID starts with **e7b**
- 2- Run the following command to get the Jenkins secret password:
docker exec -it e7b cat /var/jenkins_home/secrets/initialAdminPassword
copy the secret password
- 3- In a browser type: **localhost:8080**
Copy the secret Administrator password from the previous step and paste here:



Unlock Jenkins

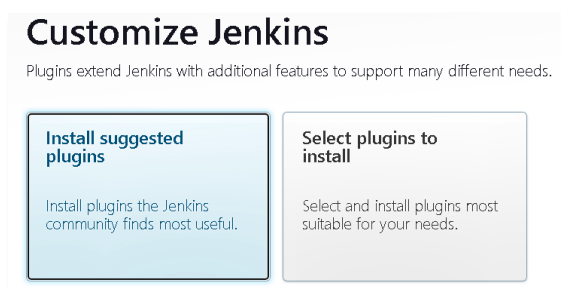
To ensure Jenkins is securely set up by the administrator, a password has been written to the log ([not sure where to find it?](#)) and this file on the server:

```
/var/jenkins_home/secrets/initialAdminPassword
```

Please copy the password from either location and paste it below.

Administrator password

- 4- Install the suggested plugins



Customize Jenkins

Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.

- 5- Create the admin user (any name and password will do. The email is not checked)

Create First Admin User

Username

Password

Confirm password

Full name

E-mail address

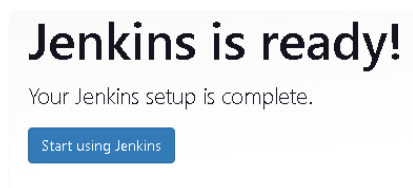
6- Follow the rest of the dialogs

Instance Configuration

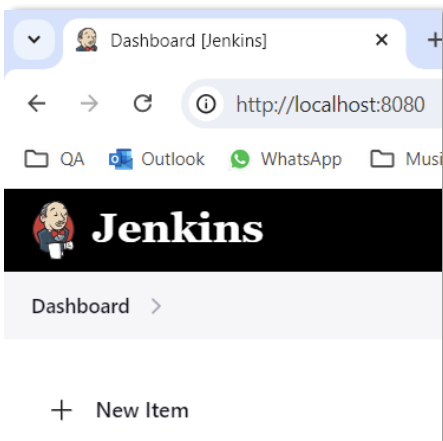
Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.



7- Create a new job by clicking [+ New Item]



8- Choose a name for this job

New Item

Enter an item name

Select an item type

**Freestyle project**
Classic general-purpose job type that checks out from up to one SCM, executes build steps serially, followed by post-build steps like archiving artifacts and sending email notifications.

9- Scroll down and find the Build Step section and then select these options:

Build Steps

Add build step ▾ ➡ Execute shell

Build Steps

 **Execute shell** ?

Command

See [the list of available environment variables](#)

```
echo build step 11
```

We will not deploy an application, only a simple script to get to know this tool.

▶ Build Now

10- Click **Save** and then

 **Build History** trend ▾

🔍 Filter... /

 **#1**
| Aug 24, 2024, 6:07 PM

 [Atom feed for all](#)  [Atom feed for failures](#)

11- Click on **#1** (build number) and then click  **Console Output**

Status

Changes

Console Output

Edit Build Information

Delete build '#1'

Timings

✓

Console Output

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/job1
[job1] $ /bin/sh -xe /tmp/jenkins5327622296918982256.sh
+ echo build step 1!
build step 1!
Finished: SUCCESS
```

12- Jenkins provides many environmental variables. We will have a look at a few. View these by clicking on the [See the list of available environment variables](#) link.

13- Click on **job1** title and then select the configure option

Configure

Execute shell ?

Command

See [the list of available environment variables](#)

echo build step 1!
echo \$BUILD_ID

▶ Build Now

14- Click

15- View the Console log and note the BUILD_ID value.

16- Modify Job1 by adding a few more lines

Add build step ▼

(Execute shell)

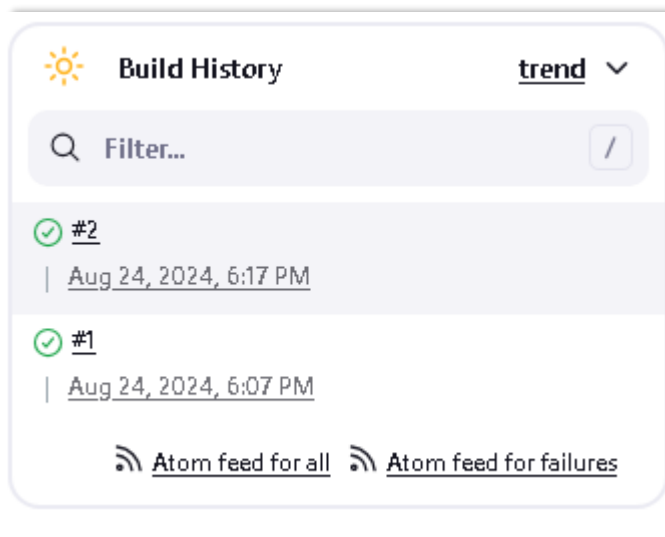
Execute shell ?

Command

See [the list of available environment variables](#)

echo builds step 2!
echo \$BUILD_NUMBER
echo \$BUILD_URL
echo \$WORKSPACE
ls -la

▶ Build Now



17- View the console log

Status

Changes

Console Output

Edit Build Information

Delete build '#3'

Timings

Previous Build

✓ Console Output

```
Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/job1
[job1] $ /bin/sh -xe /tmp/jenkins2253034970788292338.sh
+ echo build step 1!
build step 1!
+ echo 3
3
[job1] $ /bin/sh -xe /tmp/jenkins13255811510427251969.sh
+ echo builds step 2!
builds step 2!
+ echo 3
3
+ echo http://localhost:8080/job/job1/3/
http://localhost:8080/job/job1/3/
+ echo /var/jenkins_home/workspace/job1
/var/jenkins_home/workspace/job1
+ ls -la
total 8
drwxr-xr-x 2 jenkins jenkins 4096 Aug 24 18:07 .
drwxr-xr-x 3 jenkins jenkins 4096 Aug 24 18:07 ..
Finished: SUCCESS
```

18- Add another step to the build to execute a shell command to reference Jenkin's Workspace which is a directory on the Jenkins server where the files related to a specific job are stored during the build process. The following command create a file called file.txt with content of "QA Training".

Execute shell ?

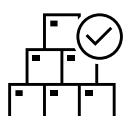
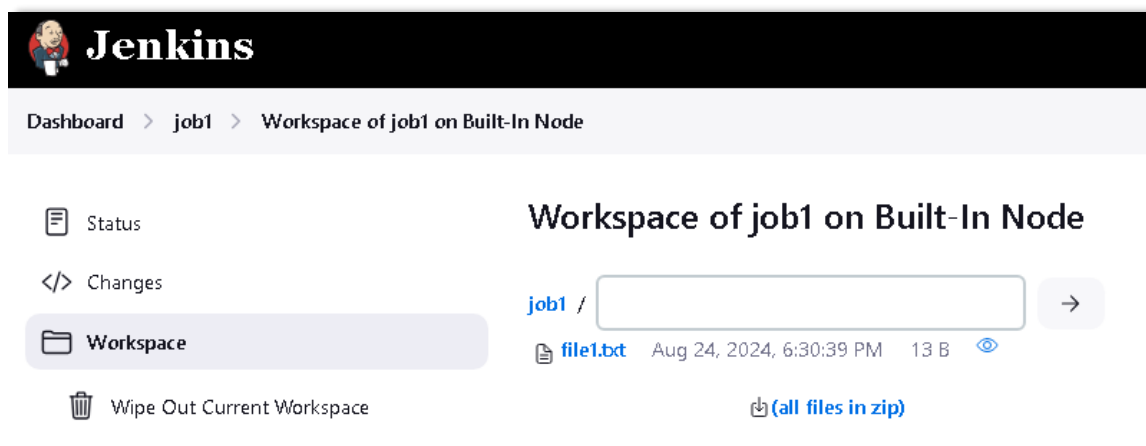
Command

See [the list of available environment variables](#)

```
echo QA Traininig > $WORKSPACE/file1.txt
```

Build Now

19- Look for the Job 1's Workspace menu and display the content of file.txt



Congratulations, you have successfully created a job using Jenkins and explored it's various IDE options.
Please continue to **Part 2** below.

Part 2- Triggers

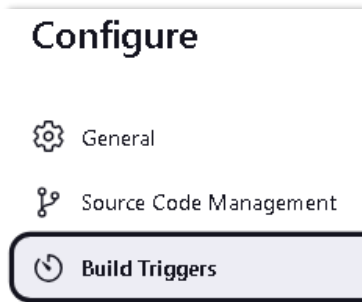
Objectives

In this part you will create a Jenkin **trigger**, and explore other forms of triggers in other labs.

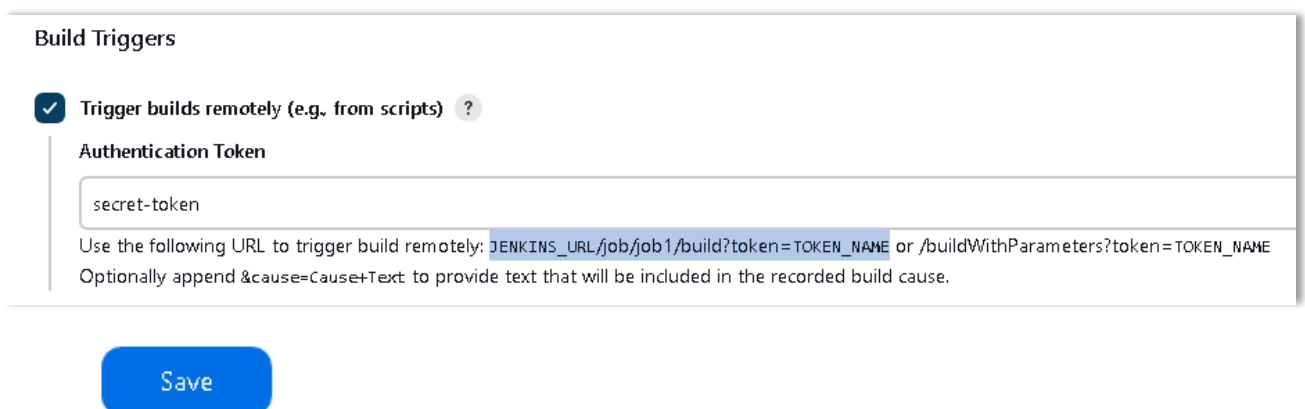
Trigger is an event or action that initiates the execution of a job or pipeline (we will explore pipelines later).

Create a Remote build trigger

- 1- Configure Job1 and find the Buil Triggers section

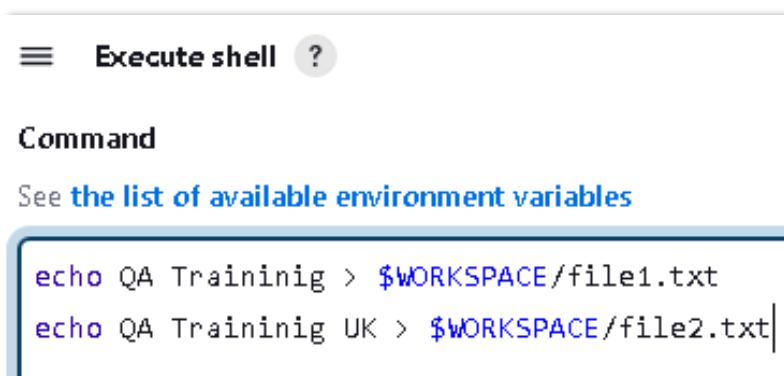


- 2- Create a token for invoking and API end point for triggering a build (any secret token word will do)



The image shows the 'Build Triggers' configuration page. The 'Trigger builds remotely (e.g. from scripts)' checkbox is checked. Below it, the 'Authentication Token' field contains the text 'secret-token'. Below the field, there is a text box with the following text: 'Use the following URL to trigger build remotely: JENKINS_URL/job/job1/build?token=TOKEN_NAME or /buildWithParameters?token=TOKEN_NAME. Optionally append &cause=Cause+Text to provide text that will be included in the recorded build cause.' At the bottom of the form is a blue 'Save' button.

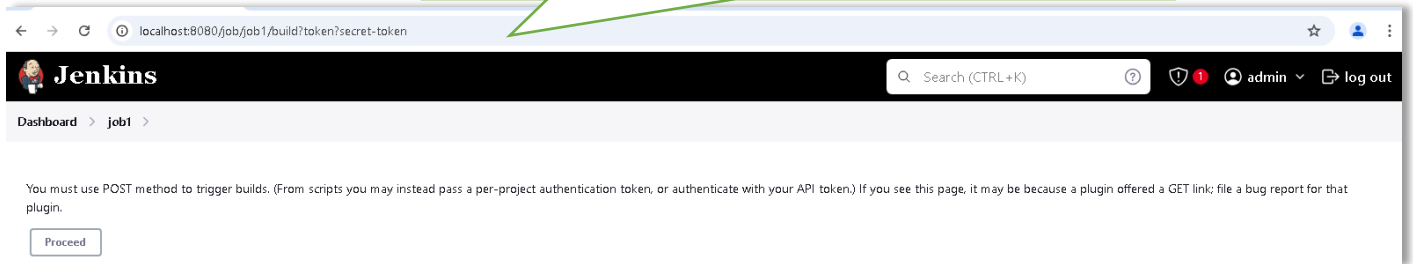
- 3- Make a change to the last build and then save.



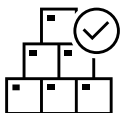
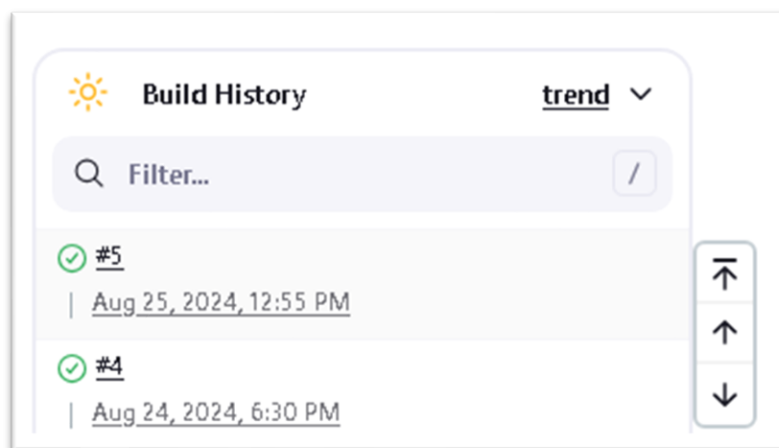
The image shows the 'Execute shell' configuration page. The 'Command' field contains two lines of shell commands: 'echo QA Traininig > \$WORKSPACE/file1.txt' and 'echo QA Traininig UK > \$WORKSPACE/file2.txt'. Above the field, there is a text box with the following text: 'See the list of available environment variables'.

- 4- Type the following text in a browser and click the Proceed button to trigger the build.
Please note, this will result in a blank screen and will only shows errors if there are any.

`http://localhost:8080/job/job1/build?token?secret-token`



- 5- As mentioned, the screen will go blank but a new build is triggered (please wait a moment until it does)



Congratulations, you have successfully created a Jenkins trigger. This is an important concept and you will explore other forms of Jenkin's Triggers in other labs.
Please continue to **Part 3** below.

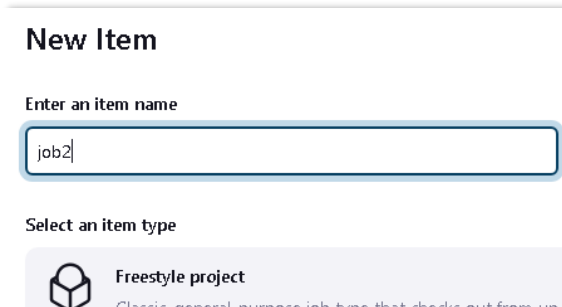
Part 3 - Chaining project builds

Objectives

In this part you will chain 2 or more builds together.

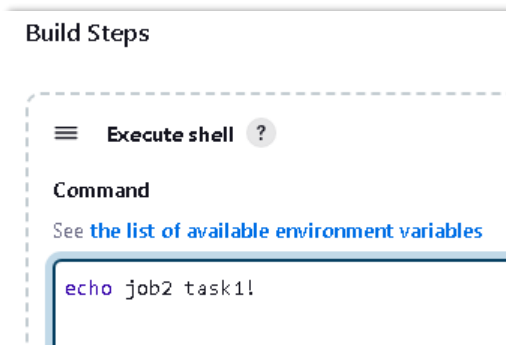
Build chaining in Jenkins automatically triggers one or more jobs after another job completes. This enables multi-step automation pipelines, where one job's output becomes the next job's input.

- 1- Create a new Job called **job2**



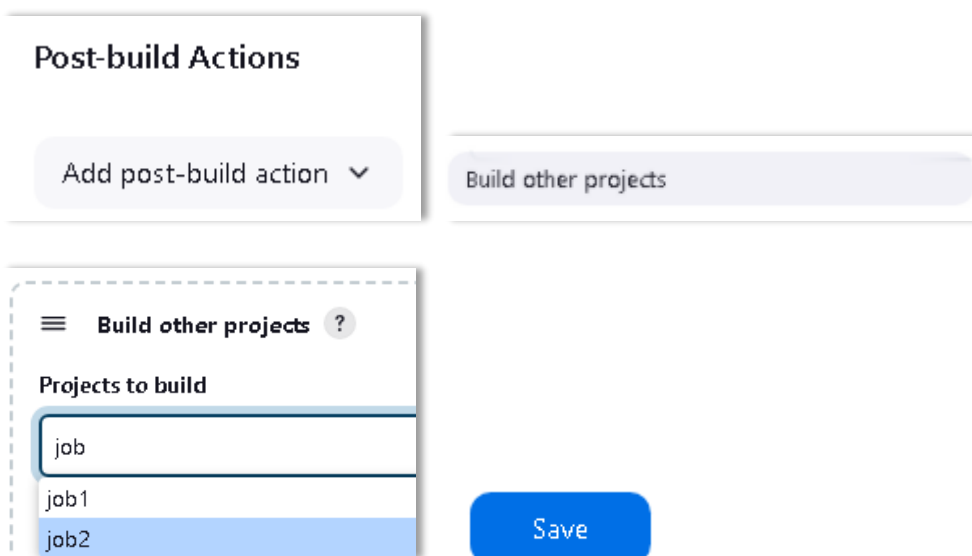
The 'New Item' form in Jenkins. It has a title 'New Item'. Below it is a text input field labeled 'Enter an item name' with the text 'job2' entered. Below that is a section 'Select an item type' with a list of options. The first option, 'Freestyle project', is selected and highlighted. Below it, a small text description reads 'Classic, general-purpose job type that checks out from un-'. There is also a small cube icon to the left of the selected option.

- 2- Assign a build step like

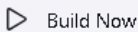


The 'Build Steps' configuration section in Jenkins. It has a title 'Build Steps'. Below it is a dashed box containing a menu icon, the text 'Execute shell', and a question mark icon. Below that is a section 'Command' with a link 'See the list of available environment variables'. At the bottom is a text input field containing the command 'echo job2 task1!'.

- 3- Configure Job1's post-build action to trigger building Job2



The 'Post-build Actions' configuration section in Jenkins. It has a title 'Post-build Actions'. Below it is a button 'Add post-build action' with a dropdown arrow. To the right is a button 'Build other projects'. Below the 'Add post-build action' button is a dashed box containing a menu icon, the text 'Build other projects', and a question mark icon. Below that is a section 'Projects to build' with a list of project names: 'job', 'job1', and 'job2'. The 'job2' option is selected and highlighted. To the right of the list is a blue button labeled 'Save'.

- 4- Build Job1  and note that both jobs get built
View the console output of Job1 and Job2 for proof

Console Output

```
Started by upstream project "job1" build number 6
originally caused by:
  Started by user admin
Running as SYSTEM
Building in workspace /var/jenkins_home/workspace/job2
[job2] $ /bin/sh -xe /tmp/jenkins3223712646613369310.sh
+ echo job2 task1!
job2 task1!
Finished: SUCCESS
```

- 5- View Job1's status for the Downstream project (Job2)

 Status

 job1

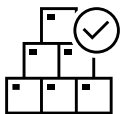
 Changes

 Workspace

 Build Now

Downstream Projects

 job2



Congratulations, you have successfully created a Jenkins build chain.

Please continue to Lab2 to explore building and deploying a Maven app stored on GIT.